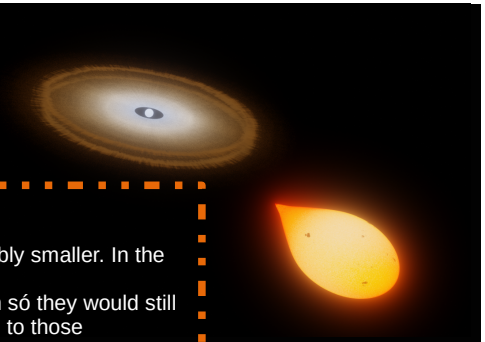
X-RAY BINARIES AT HIGH REDSHIFT

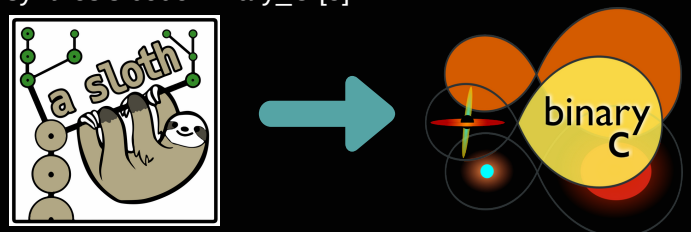


Why are XRBs important at high z?

- ***At redshifts z > 9 they dominated X-ray emission:*** as most supermassive blackholes didn't have enough time to form yet AGN numbers were considerably smaller. In the absence of AGN XRBs are the main producers of Xrays. [1]
- ***Early source of ionization of the Integalactic medium (IGM):*** X-rays mean free paths are large enough to escape primordial galaxies and small enough só they would still interact with the IGM. XRBs were able to create differences in the IGM's temperature and ionization state long before UV radiation from stars could escape to those distances. [2]
- ***Main counter-actor of Lyman-Werner radiation:*** LW destroys H2 hinders, the main coolant of primodial metal-free gas, thus hindering star formation. X-rays are able to boost H2 production by ionising H atoms and increasing the number of free electrons which catalyse H2 formation.[3]

Method

We created a catalogue of Pop III binaries usign the new semi-analytical star-formation code A-SLOTH [4] and then evolved each binary pair using population synthesis code Binary_C [5].



Results

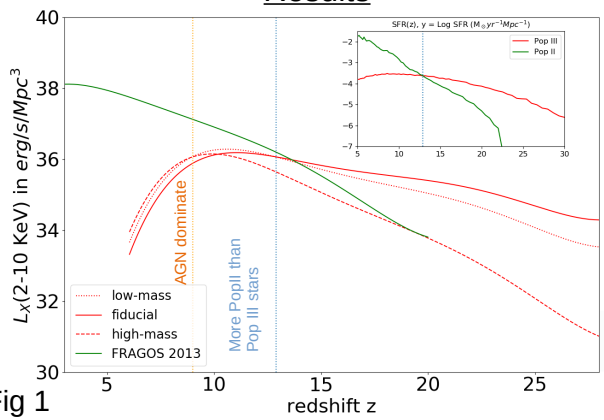


Fig 1

NAME	Min. Mass($M_\odot$)	Min. Mass($M_\odot$)	Slope (dN/dlogM)
low-mass	0.8	250	-1.35 (Salpeter)
fiducial	2	180	0.5
high-mass	10	1000	1 (flat)

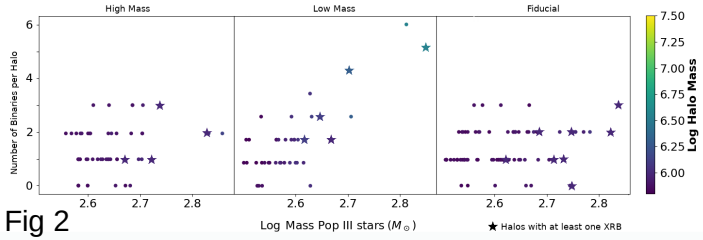


Fig 2

We used three catatogues with different IMFs (see table) to test how X-ray luminosity changes at high redshifts.
The X-ray luminosity depends on: the likelyhood of a binary forming (easier for more bottom heavy IMFs) and the ability of forming a compact object accreting from a companion (easier for top heavy IMFs). We see from Figure 1 that the low mass catalogue and the high mass catalogue both emit overall less X-rays at high redshift then the fiducial IMF. That is because, at this point, primordial galaxies are too small to create a large number of stars and the formation of XRBs becomes more IMF dependant. As we approach lower redshifts the luminosity of all three IMFs get more and more similar.

In Figure 2 we show the number of binaries in halos at redshift z = 30 and which halos possess an XRB (halos plotted as a star). At high redshifts very few halos contain an XRB which indicates that X-ray feedback was very location dependant at high redshifts. The fiducial IMF results in a higher of halos containing XRBs. The low-mass IMF allows for a larger number of stars to form given a certain stellar mass, with some halos having up to 6 binaries, but the probability of one of the binaries being an XRB is significantly decreases, whereas the high mass IMF doesn't form as many binaries to begin with limiting the number of XRBs that can form.

In the Future

In future work we will probe how different X-ray feedback affects the first generations of stars. This could have large impacts to the properties of the very first Galaxies which will be detected by the Roman Telescope. These observations will help us constrain the primordial IMFs and the XRB population in the early universe.

References

1) Fragos T., et al., 2013a, ApJ, 764, 41
2) Fialkov A., Barkana R., Visbal E., 2014, Nature, 506, 197
3) Glover S. C. O., Brand P. W. J. L., 2003, MNRAS, 340, 210
4) /www-utap.phys.s.u-tokyo.ac.jp/~hartwig/A-SLOTH.html
5) personal.ph.surrey.ac.uk/~ri0005/binary_c.html

N. Sartorio, A. Fialkov, T. Hartwing, G. Mirouh, R.Izzard ----- 05-09 October 2020 ---- Roman Conference